Term 6 Checklist

Topic	Learning Outcomes	RAG	
		1	2
Cells	Draw a labelled diagram of an animal cell including: membrane, nucleus and cytoplasm		
	Compare plant and animal cells using a diagram		
	Describe the role cell organelles e.g. membrane, nucleus, cytoplasm, mitochondria		
	Define specialised cell		
	Explain how the adaptations of specialised cells are linked to their function		
	Describe how cells make tissues and tissues make organs		
	Give three reasons why cell division takes place		
	Describe the importance of cellular respiration		
	Give the word equation for aerobic respiration		
	Describe the effects of exercise on heart and breathing rate		
	Explain why heart rate must increase during intense exercise		
Energy	Name the different forms of energy store e.g. kinetic, thermal		
	Draw energy transfer diagrams for a candle, a catapult and a TV		
	Identify useful and wasted energy in an energy transfer		
	Calculate the percentage efficiency of an energy transfer		
	Explain how fossil fuels are used to generate electricity in a power station		
	Draw a diagram showing the carbon cycle		
	Describe how human activities have affected the carbon cycle		
	Describe how carbon dioxide causes a greenhouse effect		
	Name the three most common gases in our atmosphere		
	State the approximate percentage of each gas in our atmosphere		
	Describe two processes that have caused the atmosphere to change over time		
Chemicals	Describe what an element is in terms of types of atoms		
	Compare chemical compounds to mixtures		
	Explain why chemical compounds are hard to separate		
	Describe how litmus paper can be used to test acids and bases		
	Describe how universal indicator can be used to test acids and bases		
	Match colours to numbers on the pH scale		
	Describe what a neutralisation reaction is		
	Give the chemical equation for a neutralisation reaction		
	Identify common scientific warning labels e.g. irritant, corrosive, toxic		
	Define the key terms; solution, solvent, solute and soluble		
	Draw a particle diagram to show how a solution is formed		
Forces	Describe the difference between contact and non contact forces		
	State which variable goes on each axis		
	Draw force arrows on objects that are at rest or moving		

	Calculate the resultant force of an object when two forces are acting on it		
	State the units of mass and weight		
	Explain why your weight changes on different planets but your mass does not		
	State the equation that links speed, distance and time.		
	State the units of speed, distance and time		
	Calculate speed, distance or time using the equation s = d/t. Try this one: A car travels 90m in 45s. What is its speed?		
	Describe factors that affect stopping distances		
	Describe the difference between braking distance and thinking distance		
Interdependence	Draw a food chain and web for a described feeding relationship		
	Identify producers, consumers and decomposers in a feeding relationship		
	Predict the effect of an environmental change on a food web		
	State the resources that animals and plants compete for		
	Explain why some animals are more successful competitors		
	Describe how plants can spread their seeds and why it's important		
	Describe the role of pollen and how plants reproduce		
	Explain how pollinators are useful to humans		
	Describe the role of pesticides in farming		
	Explain how bioaccumulation of pesticides occurs		
	Describe how an observational survey is conducted		
	Evaluate how useful an observational survey is		
	Describe how to measure biodiversity using a quadrat		
	Explain why biodiversity is important		
	Give the word equation for photosynthesis		
	Explain how plants are adapted for photosynthesis		
	Describe the roles of different plant structures e.g. leaves, roots, stems, flowers		
	Describe how acid rain is formed from burning fossil fuels		
	Explain how acid rain affects ecosystems		
	Explain the importance of large herbivores in rewilding projects		
	Describe what is meant by a keystone species		
	Compare vertebrates and invertebrates		
	State the general features of the five major vertebrate groups		
	Create a key which can be used to classify different species		
	Describe what it means for an animal to be adapted to its environment		
	Describe the process of evolution by natural selection		
	Compare different meanings of extinction		
	Describe different theories that explain the extinction of the dinosaurs		
Electricity	Describe how charged particles interact		
	Describe what causes static electricity		
	State which particle moves in an electrical current		

	Compare electrical conductors and insulators		
	Use a simple model to describe the difference between current and potential difference (voltage)		
	State the units of current and potential difference		
	Explain why current is the same at all points in a circuit		
	Explain why some materials conduct electricity		
	Describe the difference between a series and parallel circuit (Set 1)		
	Calculate the current in a series and parallel circuit (Set 1)		
	State how to measure current and potential difference		
	Draw complete circuit diagrams using a rule, pencil and straight lines		
	Draw a range of common circuit symbols e.g. wire, voltmeter, bulb		
	Describe the difference between AC and DC and give an example of the use of each		
	Calculate the cost to run an electrical appliance		
	Discuss how to reduce electricity costs in the home		
	State what happens when two magnets are brought close together		
	Draw a magnetic field pattern around a bar magnet		
	State the difference between a permanent magnet and an electromagnet		
	Describe the uses of electromagnets		
	Describe how to make an electromagnet stronger (use the 3cs)		
	Describe how an electric motor works		
	Evaluate the impact of electric and petrol cars on the environment		
	Describe the steps involved in how a power station works		
	Compare how a coal fired power station and wind turbine work		
	State the energy transfers carried out by solar panels		
	Give advantages and disadvantages of the use of renewable energy resources e.g. solar panels and wind turbines		
	Convert between millimetres, centimetres and metres		